

PROJECT SCOPE DOCUMENT

Asthma Care

Document Type	Project Scope
Project Name	Asthma Care
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1. Project Background

Asthma Care is a proposed patient-centered clinical support application focused on helping people with asthma access clear, evidence-grounded guidance when they have questions or symptoms. The product uses a Retrieval-Augmented Generation (RAG) approach with the selected NICE asthma guideline as its medical knowledge source.

2. Project Purpose

The purpose of Asthma Care is to provide patients with a simple way to ask asthma-related questions or describe symptoms, receive an evidence-grounded response with sources and a risk-level indication, and understand when emergency medical help is required. When emergency care is indicated, the application directs the user toward hospital care and can show nearby hospitals rather than attempting to provide treatment or medication recommendations.

3. Scope Statement

Asthma Care helps patients assess the urgency of asthma-related symptoms and access guideline-grounded information from the NICE asthma guideline, with source citations, risk-level information, emergency guidance, and nearby-hospital information when emergency care is indicated.

4. Primary User and Stakeholders

Primary User: Patient.

Secondary / Supporting User: Doctor or healthcare professional.

Healthcare Context / Stakeholder: Clinic.

The patient is the main user who interacts with the Asthma Care application. Doctors and clinics are supporting stakeholders in the wider healthcare context rather than the primary interaction target of the current prototype.

5. Clinical Scope

The current clinical scope is asthma care and asthma-related questions or symptoms that can be addressed using the selected NICE guideline.

The system focuses on:

- asthma-related questions and symptom input;
- evidence-grounded information retrieved from the NICE guideline;
- indication of the risk level of the presented situation;
- identification of situations that require emergency medical attention;
- emergency guidance directing the user to appropriate medical care; and
- nearby-hospital information for users who need emergency care.

The system is not intended to independently diagnose asthma or another medical condition.

6. Knowledge and Source Scope

The medical knowledge source for the current scope is the NICE asthma guideline used by the project. Medical answers are expected to be grounded in retrieved guideline evidence and accompanied by source information.

No patient-specific EHR or clinical-record data is included in the current scope.

7. Functional Scope — Current Product

Current patient journey:

Patient → Login → Asthma Assistant → Question / Symptoms → RAG → Answer + Sources + Risk Level → Emergency Guidance → Nearby Hospitals

Within this journey, the application is intended to:

- allow the patient to submit an asthma-related question or symptom information;
- retrieve relevant guideline evidence through the RAG system;
- provide an answer grounded in the retrieved evidence;
- show supporting sources;
- provide a risk-level indication;
- provide emergency guidance when the situation requires urgent medical attention; and
- show nearby hospitals when emergency care is indicated.

8. Explicit Exclusions

The following are outside the current product scope:

- independent medical diagnosis;
- prescribing medication;
- recommending specific medicines;
- personalized treatment plans;
- autonomous clinical decision-making;

- replacing a doctor or emergency medical service;
- providing emergency treatment instructions as a substitute for professional care;
- use of patient EHR or identifiable medical-record data; and
- medical recommendations unsupported by the selected NICE guideline.

9. Emergency-Safety Boundary

Asthma Care is not an emergency medical service. When the risk assessment indicates that emergency medical attention is required, the application should prioritize directing the user to appropriate emergency care and providing nearby-hospital information.

The application does not attempt to manage an emergency through medication or treatment recommendations. Final medical decisions and care remain the responsibility of qualified healthcare professionals.

10. System Boundary

Inside the product boundary:

- Patient login and patient interaction;
- Asthma Assistant interface;
- question/symptom input;
- RAG retrieval;
- grounded answer generation;
- source/citation presentation;
- risk-level output;
- emergency guidance; and
- nearby-hospital information.

Outside the product boundary:

- diagnosis by a medical professional;
- prescribing or dispensing medication;
- emergency clinical treatment;
- hospital clinical operations;
- patient EHR systems; and
- final clinical decision-making.

11. Current Scope vs. Future Expansion

Current Scope:

The prototype is focused on the patient-facing asthma assistant, guideline-grounded responses, source presentation, risk-level indication, emergency guidance, and nearby-hospital information.

Future Expansion:

Future versions may extend the product beyond the current prototype based on validated user and clinical needs. Potential expansion areas should be defined and approved separately before being treated as

product requirements. Future capabilities must not be presented as current functionality until implemented and validated.

12. Assumptions

- The selected NICE asthma guideline remains the authoritative medical knowledge source for the current prototype.
- Response quality depends on the coverage and quality of the indexed guideline content.
- Patients use the application as a support tool and not as a replacement for professional medical care.
- Emergency cases require professional medical attention regardless of application output.

13. Constraints

- The current medical knowledge scope is limited to the selected NICE guideline.
- The current prototype does not use patient-specific EHR data.
- The application does not prescribe medication or provide autonomous treatment decisions.
- The application is a clinical-support prototype and must maintain clear safety boundaries.

14. Scope Success Criteria

The current scope is considered satisfied when:

- a patient can access the Asthma Assistant;
- the patient can submit an asthma-related question or symptom information;
- the system retrieves relevant guideline evidence;
- the system provides a grounded answer with source information;
- the system provides a risk-level indication;
- emergency situations are directed toward appropriate medical care;
- nearby hospitals can be presented for emergency situations; and
- the system does not claim to diagnose, prescribe, or replace professional care.

15. Scope Boundary Statement

Asthma Care is a patient-centered asthma support application that uses the selected NICE asthma guideline as its medical knowledge source. Its current scope is to help patients understand asthma-related information, assess the urgency of symptoms at a high level, identify when emergency medical care is required, and locate nearby hospitals. It is not a diagnostic, prescribing, or autonomous treatment system.